

Phylogram

```
(
sequence2HBB:0.0986395
,
sequence3HBBmouse:0.0986395
):0.21122
)
```



☐ Radial

Selected **0** branches with current label

```
(
sequence1HBA 0.309859
sequence2HBB 0.0986395
sequence3HBBmouse 0.0986395
):0.09640
):0.06419;
```

Phylogram

```
(
9615.ENSACFP00000037289hund:0.29127,
(
10090.ENSMPSP00000020531Maus:0.16913,
9031.ENSALP00000012046Huhn:0.15482)
):0.09640
):0.06419;
```



☐ Radial

Selected **0** branches with current label

```
(
7955.ENSADARP00000039806zebrafish 0.39996
8030.ENSAP00000001940salmon 0.34834
9615.ENSACFP00000037289hund 0.29127
10090.ENSMPSP00000020531Maus 0.16913
9031.ENSALP00000012046Huhn 0.15482
):0.09640
):0.06419;
```

- Fill the table below

Disease	Description	Gene	Example variant	Mutation type	Functional effect
Hemochromatosis	Excess iron	HJV	NM_213653.3:c.959G>T	Missense	Likely LOF
Thalassemia	Reduced hemoglobin production leading to anemia	HBB	NC_000011.10:g.(?_5225365)(5225905_?)del	deletion	big deletion probably loss of function
Cystic Fibrosis	leading to severe damage to the lungs and digestive system due to thick, sticky mucus buildup	CFTR	NM_000492.3(CFTR):c.-893_-891delTAT	deletion	loss of function
Tay-Sachs disease	a rare, fatal, inherited neurodegenerative disorder caused by a deficiency in the HEXA enzyme, leading to toxic buildup of fatty substances (GM2 ganglioside) in the brain and spinal cord	GM2A	NM_000405.4(GM2A):c.-102A>G	single nucleotide variant	loss of function

- Above are germline variants. Find a variant that would fit oncogenicity & clinical impact for somatic variants

Disease: Liver cancer
Description: Common cancer type, ARID1A controls cell growth
Gene: ARID1A
Example variant: NM_006015:c.1273C>T
Mutation type: Nonsense
Functional effect: Loss of function / Oncogenic